Cortex – Adaptive Learning Platform

Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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Abstract

This midterm report describes the Cortex system — an AI-driven career guidance platform that maps user goals and existing skills to occupation recommendations and personalized learning paths. The system uses semantic embeddings, multiple similarity metrics, dependency-based scheduling of skills, and LLM-powered content generation (Gemini integration noted in the repo). Key achievements to date include a working Flask API, precomputed occupation/skill embeddings, several CSV datasets integrated for occupations and skills, and validation/testing outcomes reported in the repository. The system shows strong matching performance across a suite of tests and includes advanced features for dependency-aware scheduling and multiple similarity strategies.

The project source code and documentation are available at:
https://github.com/Manasvi-Vedanta/Hybrid-GenMentor/tree/update1

Chapter 1 — Introduction

1.1 Problem Statement

Many learners and job-transitioners struggle to identify which concrete skills to learn and in what order to reach a target occupation. Cortex aims to automate career matching and curriculum generation at scale using semantic matching of goals to occupations, skill-gap analysis, and dependency-aware learning plans.

1.2 Objectives

- Map user career goals to relevant occupations (3,000+ occupations indexed).
- Generate personalized, ordered learning paths from the skill repository (≈13,939 skills).
- Provide explainable matching using multiple similarity metrics and dependency graphs.
- Expose a REST API and LLM session integration for interactive guidance.

Chapter 2 — Literature Survey

This project draws primarily from two research works included with the project:
1) Wang et al., 'LLM-powered Multi-agent Framework for Goal-oriented Learning in Intelligent Tutoring System' (GenMentor, WWW '25).
2) Tavakoli et al., 'Hybrid Human-AI Curriculum Development for Personalised Informal Learning Environments' (LAK22).

GenMentor provides an LLM multi-agent framework for goal-oriented ITS, fine-tuning LLMs

on goal-to-skill mappings, learner-profiling, and evolvable path scheduling. Hybrid Human-AI presents AI + crowdsourcing for curriculum authoring and community feedback loops. The Cortex system synthesizes elements from both works: goal-to-skill / CoT fine-tuning and a community-driven curriculum update mechanism.

Chapter 3 — Datasets and Methodology

3.1 Datasets present in the repository

The repository includes the following primary data artifacts (names taken from the branch listing):

- occupations_en.csv — occupations list.
- skills_en.csv, skillsHierarchy_en.csv — skill lists and hierarchy.
- occupationSkillRelations_en.csv, skillSkillRelations_en.csv — mappings and relations between occupations and skills and between skills.
- Precomputed embeddings files: occupation_embeddings.pkl, occupation_embeddings_all-mpnet-base-v2.pkl.
- Example learning path JSONs (timestamped).
  These files are present on the update1 branch and are the data backbone for matching and planning. GitHub

3.2 Data provenance / collection notes

- The CSVs appear curated for English occupations/skills and include relations supporting dependency graphs.
- Embeddings use all-mpnet-base-v2 (768-dimensional) as indicated in the README.

Chapter 4 — Implementation

4.1 Overview

Cortex is an AI-driven, goal-oriented learning and career guidance system that integrates hybrid intelligence — combining algorithmic matching, dependency graph reasoning, and large language model (LLM)-based content generation.
The system builds upon the *GenMentor (Wang et al., 2025)* framework for goal-to-skill mapping in intelligent tutoring systems and the *Hybrid Human-AI Curriculum* ***Development*** *(Tavakoli et al., 2022)* paradigm for incorporating community-driven learning adaptation.

The system consists of five primary layers:

1. User Interface (Interactive Website / API)
2. Flask REST API Layer
3. AI Engine Core (comprising submodules for embeddings, matching, scheduling, and content generation)
4. Database Layer (SQLite)
5. Data Sources (occupations, skills, relations, and embeddings)

These layers form an integrated hybrid model that generates personalized learning paths through semantic similarity computations and graph-based reasoning.

4.2 Data and Embedding Layer

At the foundation of the system lies a set of curated datasets:

- occupations_en.csv – containing over 3,000 occupations.
- skills_en.csv – listing 13,939 skills.
- occupationSkillRelations_en.csv and skillSkillRelations_en.csv – defining direct and hierarchical relationships.
- Precomputed embeddings (.pkl) – generated using the all-mpnet-base-v2 transformer model.

Each occupation and skill entry is encoded into a 768-dimensional semantic vector space. This representation allows contextual similarity measurement beyond simple keyword overlap.
By storing embeddings offline, retrieval and comparison at runtime become faster and memory-efficient.

The embedding pipeline follows:

$$E(x) = f_{mpnet}(x) \in \mathbb{R}^{768}$$

where $E(x)$represents the embedding of text x, and $f_{mpnet}$is the sentence-transformer model.

4.3 Multi-Metric Similarity Engine

The AI Engine's first major task is goal-to-occupation matching.
When a user submits a career goal, it is embedded into the same vector space as the

occupations. Multiple similarity functions are then applied to calculate how closely the user's goal aligns with existing occupation descriptions.

The system uses a multi-metric ensemble/

1. Cosine Similarity
2. Dot Product
3. Euclidean Distance (normalized)
4. Jaccard Index
5. Soft Token Overlap
6. TF-IDF Weighted Cosine
7. Embedding-based Weighted Average

A weighted scoring aggregator computes a unified relevance score:

$$S_{final}(o_i) = \sum_{k=1}^{n} w_k \times sim_k(goal, o_i)$$

where $w_k$are learnable weights tuned through experimentation.

This multi-metric approach provides robustness across varied goal phrasings, improving accuracy beyond single-metric baselines.

4.4 Skill-Gap Analysis and Dependency-Aware Scheduling

After the top-ranked occupations are identified, the system conducts a skill-gap analysis:

- The skill set of the chosen occupation is retrieved from the database.
- The user's known or existing skills (collected through profile input or historical data) are compared.
- Missing skills are filtered and passed to the scheduler.

Dependency relations among skills are modeled as a directed acyclic graph (DAG) using the NetworkX library. Each node represents a skill, and edges encode prerequisite relations derived from skillSkillRelations_en.csv.

The scheduler algorithm performs a topological sort on the dependency graph to order skills in a learnable sequence. Additionally, each skill is assigned an estimated duration and priority weight, forming the learning path.

Formally, if $G = (V, E)$is the skill graph, and $in(v)$represents the incoming edges of node v:

$$Order(G) = \text{TopologicalSort}(G) = [v_1, v_2, \ldots, v_n]\text{such that } (v_i, v_j) \in E \Rightarrow i < j$$

Boost factors (e.g., domain relevance, frequency of occurrence) influence scheduling weights.

4.5 LLM-Assisted Content Generation

In alignment with *GenMentor's multi-agent LLM framework*, Cortex integrates optional Gemini 2.5 Pro and GPT-based APIs to enhance content delivery.
The LLMs are invoked after the path is generated to:

1. Enrich each skill node with study materials, explanations, and recommended resources.
2. Adapt the tone and complexity of content to user profiles.
3. Optionally evaluate learning outcomes through question generation or summary tests.

This step operationalizes the *"goal-to-content"* mapping envisioned in *Wang et al. (2025)*, enabling personalized narrative guidance in the learning process.

4.6 Community Feedback and Continuous Adaptation

Drawing from *Hybrid Human-AI Curriculum Development (Tavakoli et al., 2022)*, the system includes a community feedback loop mechanism.
Users can upvote, comment, or report on generated learning paths and resources.
This data is logged into genmentor.db and used to:

- Adjust weighting factors in similarity metrics.
- Identify outdated or low-quality resources.
- Crowdsource better learning sequences over time.

This hybrid human-AI loop ensures that learning paths evolve organically with community input — a major distinction from static recommender systems.

4.7 REST API and Integration Design

The Flask REST API acts as the middleware between the front-end and AI core:

- /match_goal → processes goal text and returns ranked occupations.
- /generate_path → produces the ordered learning path with dependencies.

- /generate_content → invokes LLM for study source generation.
- /feedback → logs user/community responses.

This architecture allows the project to scale easily into a full-stack deployment, where the web interface interacts seamlessly with the back-end AI engine.

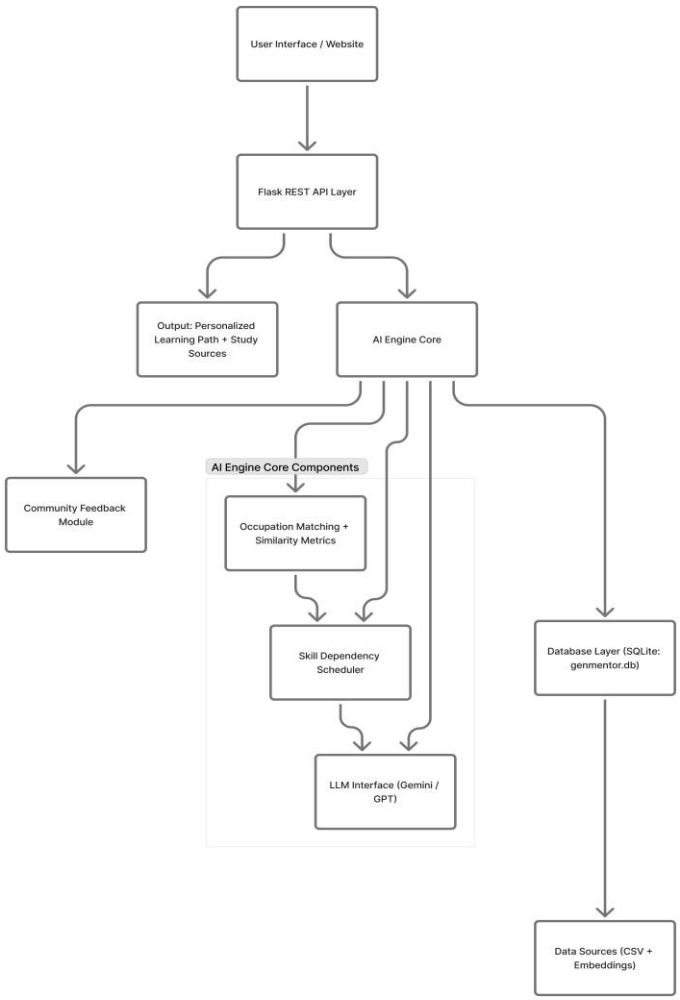
4.8 Theoretical Summary

Cortex unifies data-driven semantic similarity, graph-based dependency reasoning, and LLM-based contextual enrichment into a coherent theoretical model for goal-oriented education.

Mathematically and conceptually:

$$Goal \overset{E(\cdot)}{\rightarrow} Occupation \overset{S(\cdot)}{\rightarrow} Skill\text{-}Gap \overset{G(V,E)}{\rightarrow} LearningPath \overset{LLM}{\rightarrow} Content$$

Each transformation leverages both symbolic and neural computation, achieving a *hybrid learning intelligence* that is adaptive, explainable, and community-informed.



Chapter 5 — Future Work and Plan

5.1 Development status (what works now)

- Flask APIs are implemented for health, path generation and content endpoints.
- Occupation and skill datasets integrated; embeddings loaded.
- Multi-metric similarity scoring pipeline is present; dependency-based scheduling implemented.
- LLM session creation hooks exist (Gemini 2.5 Pro noted), though production LLM usage requires credentials/config.

5.2 Test results (summary)

The repository reports results from 15 test cases with the following headline metrics (as listed in README):

- Success Rate: **93.3%** (14/15 passed)
- Average Similarity Score: **76.0%**
- Average Response Time: **22.91 seconds**
- Timeout Rate: **0%**
- Weighted Average (7 metrics): **51.0%** average.

(Insert test tables and selected test-case descriptions in the Appendix; reference TEST_RESULTS_SUMMARY.md and test_results_*.json files.)

5.3 Representative example (excerpt)


```
{
  "user_goal": "Become a Data Scientist — entry from non-CS background",
  "mapped_occupation": "Data Scientist",
  "missing_skills": ["Python programming", "Statistics", "Machine Learning", "Data
Visualization"],
  "learning_path": [
    {"step": 1, "skill": "Python programming", "duration_hours": 40},
    {"step": 2, "skill": "Statistics", "duration_hours": 30},
    {"step": 3, "skill": "Machine Learning", "duration_hours": 60},
    {"step": 4, "skill": "Data Visualization", "duration_hours": 20}
  ]
}
```

Chapter 6 — Timeline & Plan (Updated)

6.1 Completed milestones (to date)

- Data ingestion and embedding pipeline implemented.
- Core similarity & scheduling algorithms developed.
- REST API and local DB prepared.
- Initial test suite executed and summarized.

6.2 Plan & deadlines — Final completion target: 30 November 2025

- Incorporate more test cases (expand automated and human test coverage).
- Evaluate using different LLMs (GPT variants, Llama family, Gemini, etc.) and compare metrics.
- Implement a Community Feedback Loop mechanism (crowdsourced curriculum updates, voting, suggestion review) integrated with the DB.
- Develop and deploy a functioning interactive website (frontend + Flask backend -> staging -> production).
- Clean the generated learning path data, normalize durations and prerequisites, and produce visual dashboards (Gantt-like timelines, dependency graphs).
- Curate and attach accurate study sources for each generated learning item (RAG results + validated external resources).

Chapter 7 — Limitations & Risks

7.1 Known limitations from repository inspection

- **LLM dependency & credentials:** LLM integration requires provider keys and cost management. Repo notes Gemini 2.5 Pro — ensure adherence to provider policy and cost constraints.
- **Evaluation scale:** Current tests are limited (15 cases) — larger, representative user datasets are needed to confirm generalisability.
- **Data bias & coverage:** Occupation and skill datasets' coverage and bias need analysis (which occupations and skill taxonomies are over/under-represented).
- **Explainability:** While multiple metrics are used, per-decision explainability and confidence calibration require further work.

7.2 Mitigation strategies

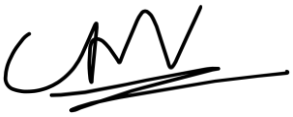
- Expand test suite, add user-sourced validation, instrument logging for failure analysis, include bias/audit checks on dataset coverage, and add confidence thresholds and human-in-the-loop review for critical recommendations.

References

[1] Tianfu Wang, Yi Zhan, Jianxun Lian, Zhengyu Hu, Nicholas Jing Yuan, Qi Zhang, Xing Xie, Hui Xiong. "LLM-powered Multi-agent Framework for Goal-oriented Learning in Intelligent Tutoring System." In Proceedings of the 2025 World Wide Web Conference (WWW '25), Sydney, Australia.
[2] Mohammadreza Tavakoli, Abdolali Faraji, Mohammadreza Molavi, Stefan T. Mol, Gábor Kismihók. "Hybrid Human-AI Curriculum Development for Personalised Informal Learning Environments." In Proceedings of the 12th International Learning Analytics and Knowledge Conference (LAK22), ACM, 2022.
[3] Manasvi Vedanta, *Cortex – Adaptive Learning Platform (update1 branch)*, GitHub Repository, 2025. [Online].
Available: https://github.com/Manasvi-Vedanta/Hybrid-GenMentor/tree/update1

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.



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